TRIBHUVAN UNIVERSITY

INSTITUTE OF ENGINEERING

PULCHOWK CAMPUS

Mini-Project: Network Design

Submitted By:

Shrine Sigdel (079BCT080)

Submitted To:

Assoc. Prof. Sharad Kumar Ghimire

Computer Networks

Department of Electronics & Computer Engineering

August, 2026

1 Introduction

This report documents the completed design and Cisco Packet Tracer implementation of an enterprise network for KtmTechnologies, a technology and electronics organization with a corporate headquarters, an R&D department, a production and logistics plant, a data center and a branch sales office. The network is built around a three-area OSPF hierarchy (Area 0 backbone, Area 1, Area 2), grouping departmental LANs and VLANs across ten internal routers connected by fourteen dedicated point-to-point links, with a redundant core (CR1/CR2) providing multiple paths into two of the areas.

IP addressing was planned using Variable Length Subnet Masking (VLSM) on the private block 10.0.0.0/16. This report retains the original OSPF area plan and VLSM addressing approach, and documents the implementation details finalized during configuration: VLAN ID assignments, DHCP server placement and pool ranges, and a hierarchical, recursive-style internal/external DNS and web-hosting configuration.

2 Network Topology

2.1 OSPF Area Plan

Area	Member Networks
Area 0 (Backbone)	Border Router (BR), Core Routers (CR1, CR2) — transit only, Data Center Server LAN (R11)
Area 1	HQ Corporate VLANs — Finance, HR, Legal (R4); Sales, Marketing (R8)
Area 2	R&D Lab (R5), Production Floor (R6), Warehouse & Logistics (R7), Branch Office (R9)

Table 1: OSPF Area Plan

2.2 Router and LAN Summary

Device	Area	Router Connections	LAN(s) Served
ISP-R1	ISP	BR (P01)	ISP DNS LAN, ISP simulated external LAN
BR	0	ISP, CR1, CR2 (P01–P03)	None; transit router
CR1	0	BR, CR2, R11, R5, R4	None; transit/core router
CR2	0	BR, CR1, R11, R5, R4, R6	None; transit/core router
R11	0	CR1, CR2 (P05, P06)	Data Center Server LAN
R4	1 (ABR)	CR1, CR2, R8 (P09–P11)	Finance, HR, Legal VLANs
R8	1	R4 (P11)	Sales, Marketing

Device	Area	Router Connections	LAN(s) Served
R5	2 (ABR)	CR1, CR2 (P07, P08)	R&D Lab
R6	2 (ABR)	CR2, R7 (P12, P13)	Production Floor
R7	2	R6, R9 (P13, P14)	Warehouse & Logistics
R9	2	R7 (P14)	Branch Office

Table 2: Router and LAN Summary

2.3 Logical Topology in Cisco Packet Tracer

The proposed topology has been fully built and configured in Cisco Packet Tracer. Each departmental LAN and VLAN segment has a corresponding router, switch and set of end devices, connected through the planned point-to-point backbone links. End devices (PCs, servers) were added per LAN during final configuration, and interfaces were addressed according to the VLSM plan below.

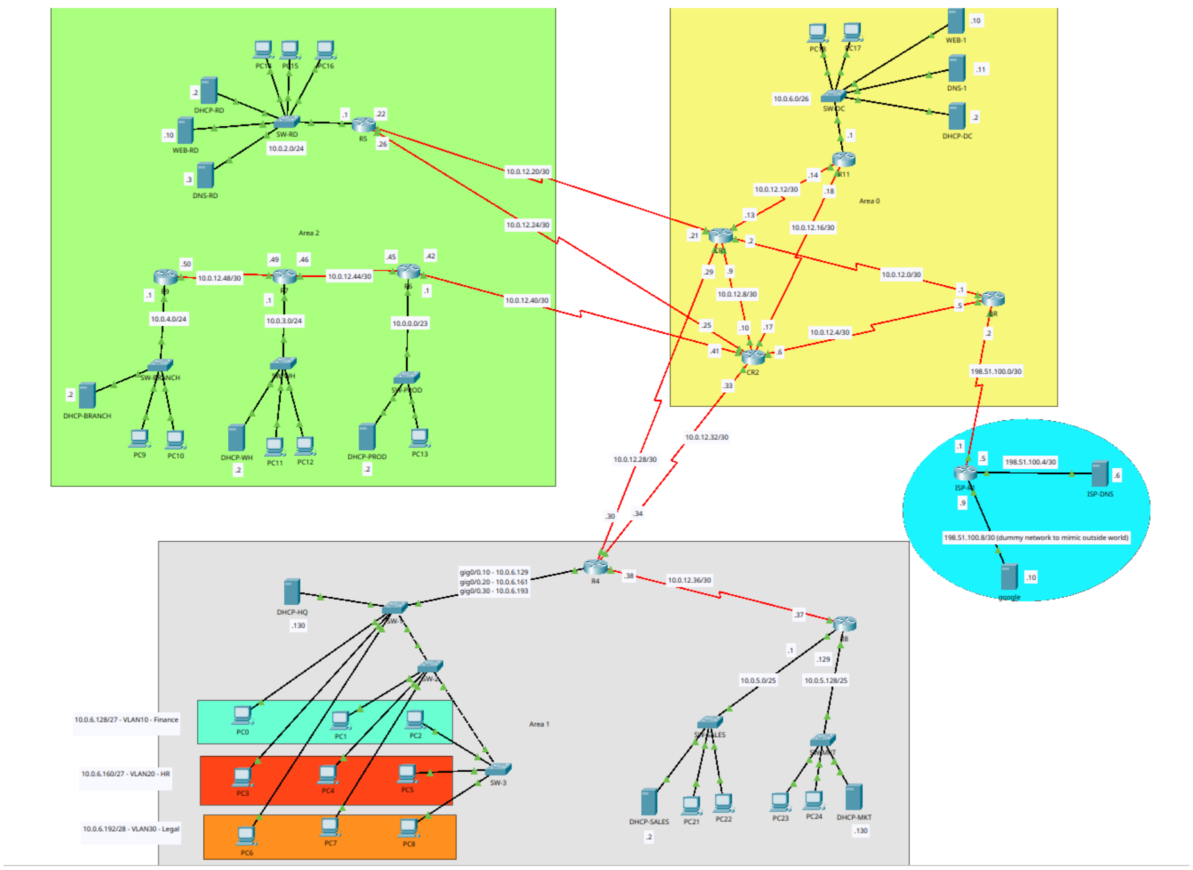

Figure 1: Full network topology (Cisco Packet Tracer)

3 Addressing Description

3.1 IP Address Pool

- Private enterprise block: 10.0.0.0/16
- Allocated LAN/VLAN range: 10.0.0.0 – 10.0.6.207
- Internal point-to-point pool: 10.0.12.0/24
- Reserved growth space: 10.0.6.64/26 (for Data Centre , area 0), 10.0.7.0 – 10.0.11.255, and 10.0.13.0 onward
- ISP documentation blocks (RFC 5737): 198.51.100.0/30 (BR–ISP link), 198.51.100.4/30 (ISP DNS LAN), 198.51.100.8/30 (simulated external/"outside world" network hosting a simulated google.com web server, used to demonstrate the full recursive DNS chain)

3.2 Subnets — VLSM Table

Ten LAN/VLAN subnets in total, using six prefix sizes (/23, /24, /25, /26, /27, /28), allocated largest to smallest to reduce fragmentation.

Sn	Network	Req.	Alloc.	Mask	Net. ID	Usable Range	VLAN
1	Production Floor	450	510	255.255.254.0	10.0.0.0/23	.0.1–.1.254	—
2	R&D Lab	240	254	255.255.255.0	10.0.2.0/24	.2.1–.2.254	—
3	Warehouse & Logistics	200	254	255.255.255.0	10.0.3.0/24	.3.1–.3.254	—
4	Branch Office	150	254	255.255.255.0	10.0.4.0/24	.4.1–.4.254	—
5	Sales	100	126	255.255.255.128	10.0.5.0/25	.5.1–.5.126	—
6	Marketing	80	126	255.255.255.128	10.0.5.128/25	.5.129–.5.254	—
7	Data Center / Server LAN	40	62	255.255.255.192	10.0.6.0/26	.6.1–.6.62	—
8	Finance VLAN	20	30	255.255.255.224	10.0.6.128/27	.6.129–.6.158	10
9	HR VLAN	20	30	255.255.255.224	10.0.6.160/27	.6.161–.6.190	20
10	Legal VLAN	9	14	255.255.255.240	10.0.6.192/28	.6.193–.6.206	30

Table 3: VLSM subnet allocation

3.3 Point-to-Point Router Links

14 inter-router links (P01–P14), including 1 ISP link, each using a /30 subnet. Redundant links (P05/P06, P07/P08, P09/P10) provide dual paths from the core into the Data Center, R&D and HQ

areas. All point-to-point links are implemented on serial interfaces (e.g. Serial0/2/0, Serial0/3/0, Serial0/3/1).

Link	Network ID	Rtr A	IP A	Rtr B	IP B	Purpose
P01	198.51.100.0/30	ISP-R1	.1	BR	.2	ISP–BR uplink
P02	10.0.12.0/30	BR	.1	CR1	.2	BR–CR1
P03	10.0.12.4/30	BR	.5	CR2	.6	BR–CR2
P04	10.0.12.8/30	CR1	.9	CR2	.10	Core loop
P05	10.0.12.12/30	CR1	.13	R11	.14	CR1–R11
P06	10.0.12.16/30	CR2	.17	R11	.18	CR2–R11 (redundant)
P07	10.0.12.20/30	CR1	.21	R5	.22	CR1–R5
P08	10.0.12.24/30	CR2	.25	R5	.26	CR2–R5 (redundant)
P09	10.0.12.28/30	CR1	.29	R4	.30	CR1–R4
P10	10.0.12.32/30	CR2	.33	R4	.34	CR2–R4 (redundant)
P11	10.0.12.36/30	R8	.37	R4	.38	R4–R8
P12	10.0.12.40/30	CR2	.41	R6	.42	CR2–R6
P13	10.0.12.44/30	R7	.46	R6	.45	R6–R7 plant backbone
P14	10.0.12.48/30	R9	.50	R7	.49	R7–R9 branch link

Table 4: Point-to-point link addressing (as configured)

4 VLAN Design and Assignment

The HQ section (Finance, HR and Legal) required more than one logically separated network on the same physical link, so VLANs with router-on-a-stick inter-VLAN routing were configured on R4.

Group	VLAN Name	ID	Subnet	Router / Sub-interface
HQ	Finance	10	10.0.6.128/27	R4, Gi0/0.10 — 10.0.6.129
HQ	HR	20	10.0.6.160/27	R4, Gi0/0.20 — 10.0.6.161
HQ	Legal	30	10.0.6.192/28	R4, Gi0/0.30 — 10.0.6.193

Table 5: VLAN ID assignment (R4, router-on-a-stick)

Figure 2 shows the three HQ departmental VLANs in Packet Tracer. The Finance, HR and Legal segments correspond respectively to VLAN IDs 10, 20 and 30 listed in Table 5.

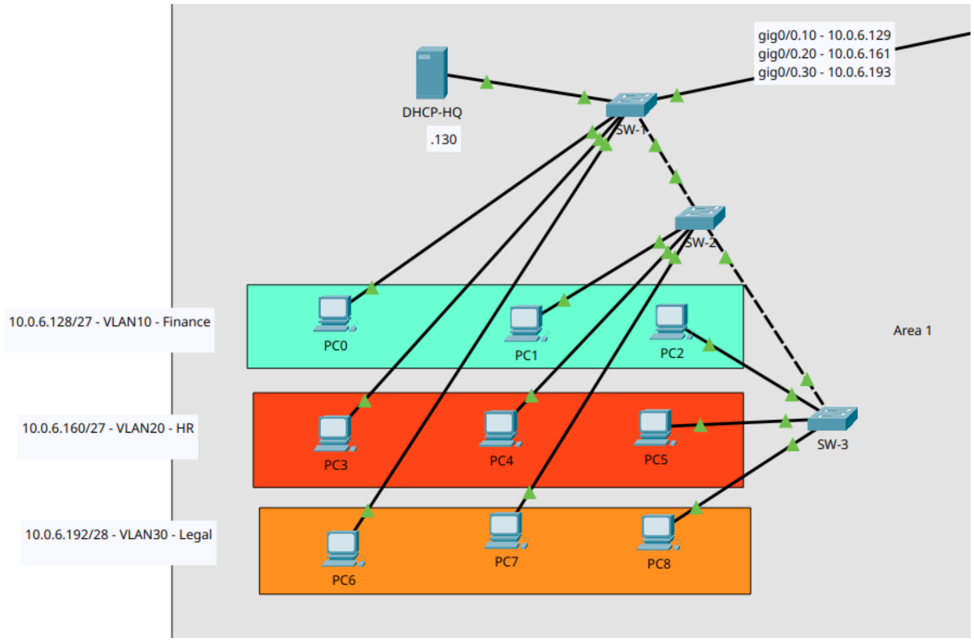

Figure 2: HQ Finance, HR and Legal VLAN implementation

Each sub-interface uses 802.1Q encapsulation matching its VLAN ID, and the switch port facing R4 is configured as a trunk carrying VLANs 10, 20 and 30.

5 DHCP Configuration

Every LAN/VLAN uses the first usable address of its subnet as the default gateway. Standalone LANs (Production, R&D, Warehouse, Branch, Sales, Marketing, Data Center) each host a local DHCP server. The three HQ VLANs (Finance, HR, Legal) share a single physical DHCP server hosted on the Finance VLAN, which serves the HR and Legal VLANs via DHCP relay (ip helper-address) configured on the corresponding R4 sub-interfaces.

LAN / VLAN	Subnet	Gateway	DHCP Server	Pool Start
Production Floor	10.0.0.0/23	10.0.0.1	10.0.0.2 (local)	10.0.0.10
R&D Lab	10.0.2.0/24	10.0.2.1	10.0.2.2 (local)	10.0.2.11
Warehouse & Logistics	10.0.3.0/24	10.0.3.1	10.0.3.2 (local)	10.0.3.10
Branch Office	10.0.4.0/24	10.0.4.1	10.0.4.2 (local)	10.0.4.10
Sales	10.0.5.0/25	10.0.5.1	10.0.5.2 (local)	10.0.5.10
Marketing	10.0.5.128/25	10.0.5.129	10.0.5.130 (local)	10.0.5.135
Data Center	10.0.6.0/26	10.0.6.1	10.0.6.2 (local)	10.0.6.15
Finance (VLAN 10)	10.0.6.128/27	10.0.6.129	10.0.6.130 (local)	10.0.6.131

LAN / VLAN	Subnet	Gateway	DHCP Server	Pool Start
HR (VLAN 20)	10.0.6.160/27	10.0.6.161	10.0.6.130 (relayed)	10.0.6.162
Legal (VLAN 30)	10.0.6.192/28	10.0.6.193	10.0.6.130 (relayed)	10.0.6.194

Table 6: DHCP server placement and pool configuration

Figure 3 shows the DHCP-HQ pool configuration, including its default gateway, DNS server and starting address. These visible settings confirm that the Finance-VLAN-hosted DHCP-HQ server matches the values documented in Table 6 and supplies the relayed HQ scopes.

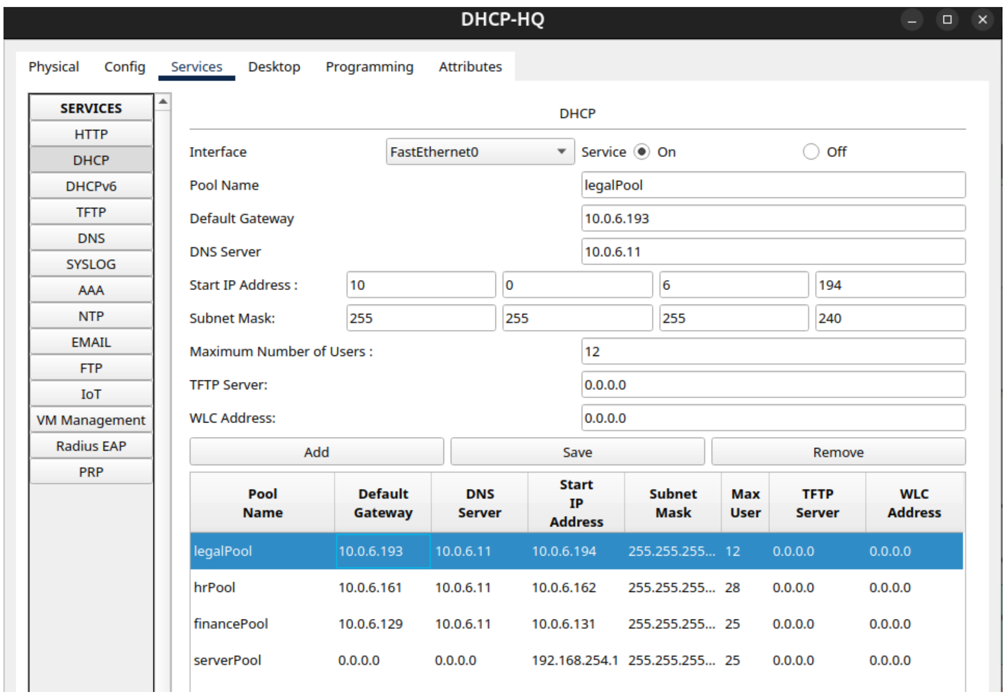

Figure 3: DHCP-HQ server pool configuration

6 Servers, DNS and Web Hosting

The Data Center (Area 0) hosts the company's primary website and primary DNS resolver. The R&D Lab (Area 2) hosts a second, access-restricted research website and a secondary DNS resolver. On the ISP side, a top-level DNS server and a simulated external website represent the outside world for testing recursive name resolution and internet-bound traffic.

Server	IP	Subnet / Location	Domain / Description
Web-1	10.0.6.10	10.0.6.0/26, Data Center, Area 0	www.ktmtech.com — primary corporate website

Server	**IP**	**Subnet / Location**	**Domain / Description**
DNS-1	10.0.6.11	10.0.6.0/26, Data Center, Area 0	Resolves www.ktmtech.com directly; delegates all other .com queries to ISP-DNS
DHCP-DC	10.0.6.2	10.0.6.0/26, Data Center, Area 0	Issues dynamic IPs to the Data Center LAN
Web-RD	10.0.2.10	10.0.2.0/24, R&D Lab, Area 2	www.ktmresearchprivate.com — private research site
DNS-RD	10.0.2.3	10.0.2.0/24, R&D Lab, Area 2	Resolves the private research domain directly; delegates all other .com queries to DNS-1
DHCP-RD	10.0.2.2	10.0.2.0/24, R&D Lab, Area 2	Issues dynamic IPs to the R&D LAN
ISP-DNS	198.51.100.6	198.51.100.4/30, ISP LAN	Top-level resolver; holds no KtmTechnologies records; answers for simulated external domains (e.g. google.com)
Google-Sim (Web)	198.51.100.10	198.51.100.8/30, ISP "outside world" LAN	www.google.com (simulated) — represents an external site reachable only via full recursive delegation

Table 7: Server, DNS and web-hosting addresses

The three browser tests in Figures 4–6 were all performed from PC15 in the R&D network (Area 2). Together they demonstrate direct resolution by DNS-RD, delegation from DNS-RD to DNS-1, and the complete DNS-RD–DNS-1–ISP-DNS resolution chain.

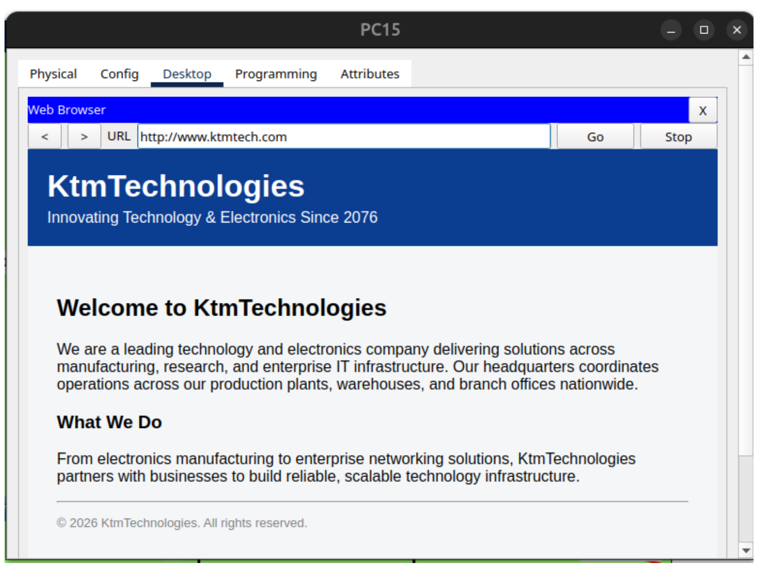

Figure 4: PC15 accessing www.ktmtech.com through DNS-RD to DNS-1 delegation

Figure 5 confirms that PC15 can load the private research website through a direct match on its local DNS-RD server, without further delegation.

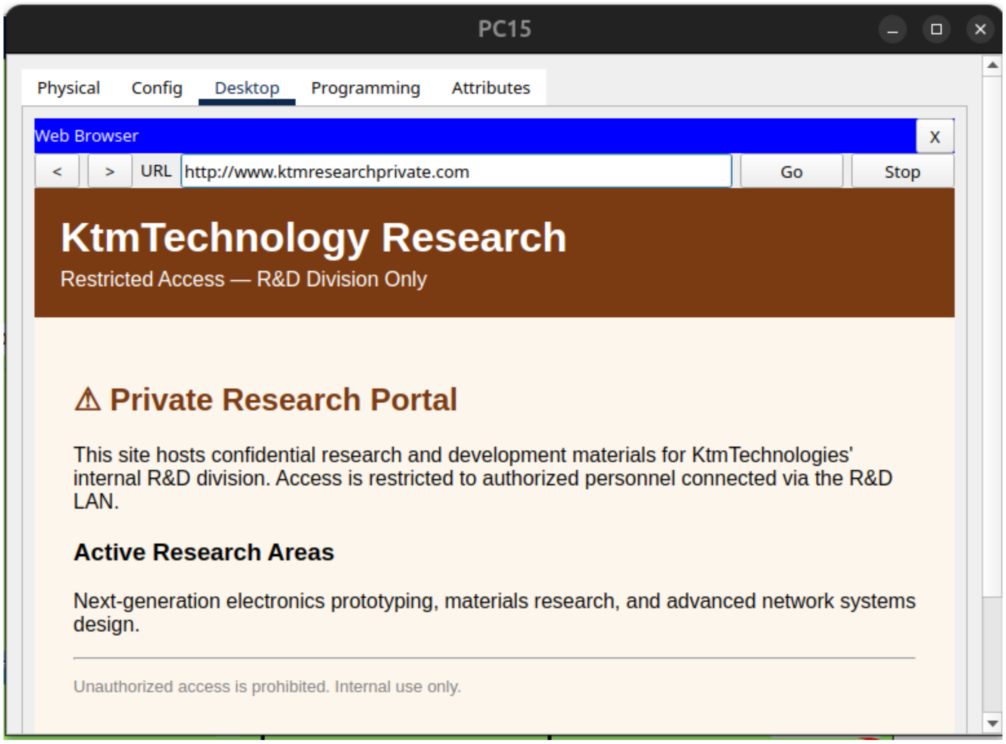

Figure 5: PC15 accessing the private R&D website through direct DNS-RD resolution

Figure 6 demonstrates successful external-style resolution and web access from the same client using the full three-server delegation path.

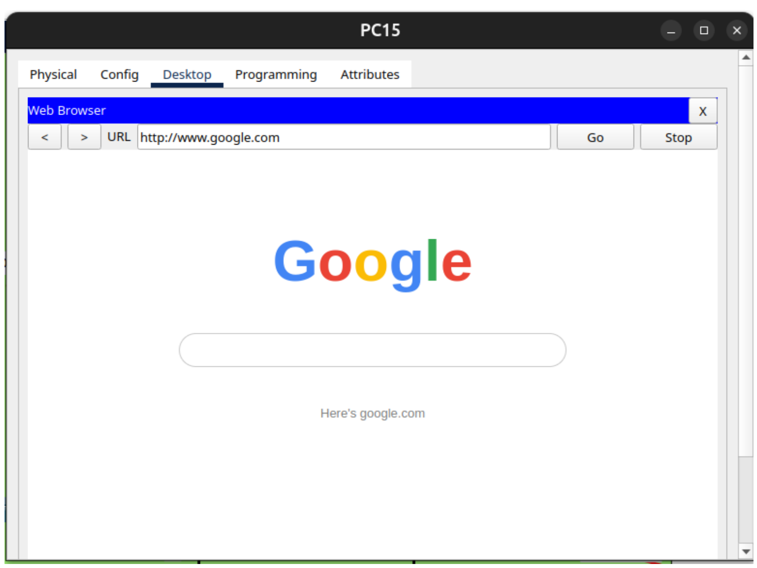

Figure 6: PC15 accessing simulated www.google.com through the full DNS delegation chain

6.1 Recursive DNS Resolution Design

Rather than a single flat DNS record table, name resolution is structured as a hierarchical, recursive-style chain using Packet Tracer's NS-record delegation (an NS record referencing a local "glue" A record that resolves to the next server in the chain):

- DNS-RD holds only its own domain (www.ktmresearchprivate.com → 10.0.2.10) plus an NS delegation for the generic .com zone pointing to DNS-1 (10.0.6.11).
- DNS-1 holds only its own domain (www.ktmtech.com → 10.0.6.10) plus an NS delegation for the generic .com zone pointing to ISP-DNS (198.51.100.6).
- ISP-DNS holds no KtmTechnologies records at all — it is the top of the chain and would answer only for domains genuinely outside company control, such as the simulated google.com record pointing to 198.51.100.10.

Figure 7 shows DNS-1's A record for www.ktmtech.com together with the NS delegation and supporting address record that direct other .com queries toward ISP-DNS. These visible records implement the middle stage of the delegation chain described above.

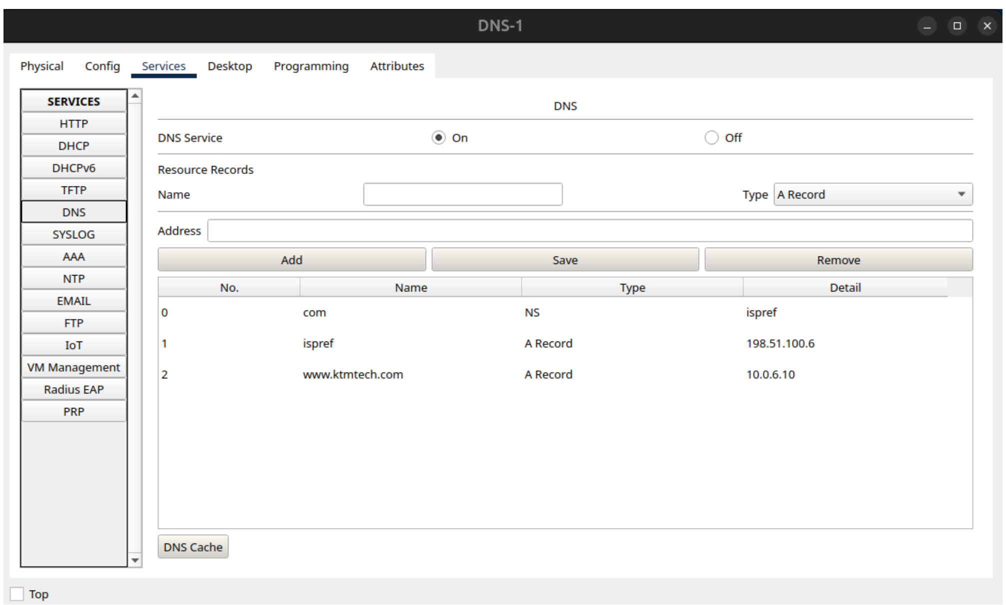

Figure 7: DNS-1 record table and delegation toward ISP-DNS

Figure 8 shows DNS-RD's direct A record for the private research domain and its NS delegation, with the associated address record, toward DNS-1. This configuration forms the first stage used by Area 2 clients when a requested name is not resolved locally.

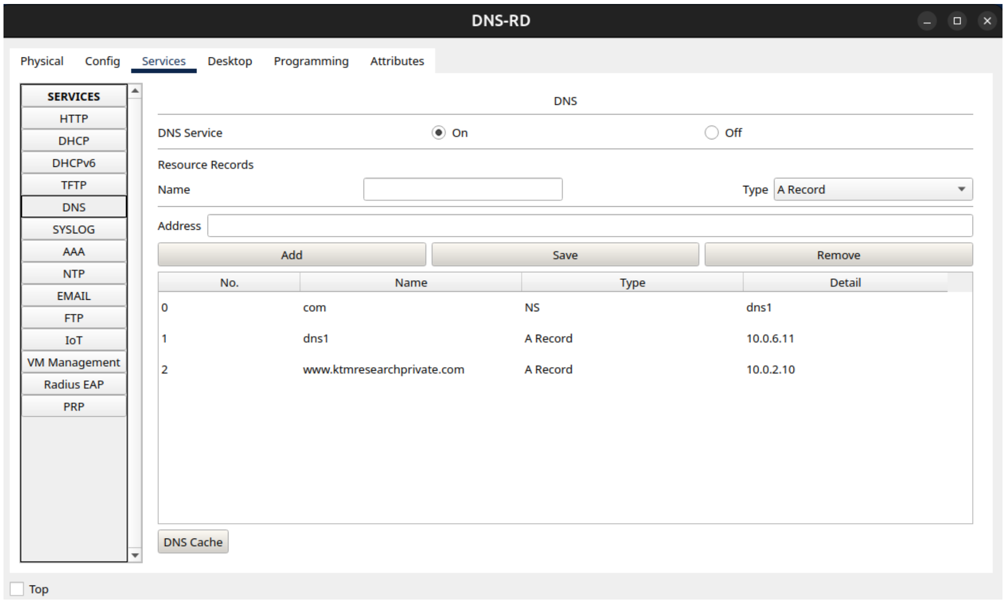

Figure 8: DNS-RD record table and delegation toward DNS-1

Each area's DHCP scope assigns its own "local" resolver: Area 0 and Area 1 clients use DNS-1; Area 2 (R&D) clients use DNS-RD. This produces two practical effects worth noting for the report:

- **Resolution chain example** — an R&D client browsing www.google.com: query goes to DNS-RD → no match → delegated to DNS-1 → no match → delegated to ISP-DNS → answered (198.51.100.10).
- **Resolution chain example** — an R&D client browsing its own private site: query goes to DNS-RD → matched directly (10.0.2.10), no delegation needed.
- **Access-restriction effect** — a Data Center or HQ client (using DNS-1) cannot resolve www.ktmresearchprivate.com, since DNS-1 has no delegation back to DNS-RD for that name. This is a resolution-time restriction arising from DNS scope, not an access-control list, and is noted as a design limitation rather than a security guarantee.

6.2 Routing Verification

Figure 9 records the `show ip route` output from BR (Shrine1). It confirms that BR has routes to all ten LAN/VLAN subnets through OSPF while retaining a single static default route toward the ISP. This verifies the minimum-route-entry design described in Section 7: internal reachability is learned dynamically, while only the necessary ISP-facing default is configured statically on BR.

```
Shrine1>show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

Gateway of last resort is 198.51.100.1 to network 0.0.0.0

     10.0.0.0/8 is variably subnetted, 25 subnets, 8 masks
O IA    10.0.0.0/23 [110/129] via 10.0.12.6, 00:38:17, Serial0/3/1
O IA    10.0.2.0/24 [110/129] via 10.0.12.2, 00:38:17, Serial0/2/0
                    [110/129] via 10.0.12.6, 00:38:17, Serial0/3/1
O IA    10.0.3.0/24 [110/193] via 10.0.12.6, 00:38:17, Serial0/3/1
O IA    10.0.4.0/24 [110/257] via 10.0.12.6, 00:38:17, Serial0/3/1
O IA    10.0.5.0/25 [110/193] via 10.0.12.6, 00:38:17, Serial0/3/1
                    [110/193] via 10.0.12.2, 00:38:17, Serial0/2/0
O IA    10.0.5.128/25 [110/193] via 10.0.12.6, 00:38:17, Serial0/3/1
                      [110/193] via 10.0.12.2, 00:38:17, Serial0/2/0
O       10.0.6.0/26 [110/129] via 10.0.12.2, 00:38:17, Serial0/2/0
                    [110/129] via 10.0.12.6, 00:38:17, Serial0/3/1
O IA    10.0.6.128/27 [110/129] via 10.0.12.6, 00:38:17, Serial0/3/1
                      [110/129] via 10.0.12.2, 00:38:17, Serial0/2/0
O IA    10.0.6.160/27 [110/129] via 10.0.12.6, 00:38:17, Serial0/3/1
                      [110/129] via 10.0.12.2, 00:38:17, Serial0/2/0
O IA    10.0.6.192/28 [110/129] via 10.0.12.6, 00:38:17, Serial0/3/1
                      [110/129] via 10.0.12.2, 00:38:17, Serial0/2/0
C       10.0.12.0/30 is directly connected, Serial0/2/0
L       10.0.12.1/32 is directly connected, Serial0/2/0
C       10.0.12.4/30 is directly connected, Serial0/3/1
L       10.0.12.5/32 is directly connected, Serial0/3/1
O       10.0.12.8/30 [110/128] via 10.0.12.2, 00:38:17, Serial0/2/0
                     [110/128] via 10.0.12.6, 00:38:17, Serial0/3/1
O       10.0.12.12/30 [110/128] via 10.0.12.2, 00:38:17, Serial0/2/0
O       10.0.12.16/30 [110/128] via 10.0.12.6, 00:38:17, Serial0/3/1
O IA    10.0.12.20/30 [110/128] via 10.0.12.2, 00:38:17, Serial0/2/0
O IA    10.0.12.24/30 [110/128] via 10.0.12.6, 00:38:17, Serial0/3/1
O IA    10.0.12.28/30 [110/128] via 10.0.12.2, 00:38:17, Serial0/2/0
O IA    10.0.12.32/30 [110/128] via 10.0.12.6, 00:38:17, Serial0/3/1
O IA    10.0.12.36/30 [110/192] via 10.0.12.6, 00:38:17, Serial0/3/1
                      [110/192] via 10.0.12.2, 00:38:17, Serial0/2/0
O IA    10.0.12.40/30 [110/128] via 10.0.12.6, 00:38:17, Serial0/3/1
O IA    10.0.12.44/30 [110/192] via 10.0.12.6, 00:38:17, Serial0/3/1
O IA    10.0.12.48/30 [110/256] via 10.0.12.6, 00:38:17, Serial0/3/1
     198.51.100.0/24 is variably subnetted, 2 subnets, 2 masks
C       198.51.100.0/30 is directly connected, Serial0/3/0
L       198.51.100.2/32 is directly connected, Serial0/3/0
S*   0.0.0.0/0 [1/0] via 198.51.100.1
```

Figure 9: BR (Shrine1) routing table verification using `show ip route`

7 Additional Specifications

- **Addressing scheme:** VLSM on 10.0.0.0/16, subnets allocated largest-host-count first to minimize fragmentation.
- **Routing protocol:** OSPFv2, multi-area (Area 0 backbone + Areas 1–2), with R4, R5 and R6 acting as Area Border Routers connecting their areas to the backbone.
- **VLANs:** Inter-VLAN routing performed via router-on-a-stick on R4 for the Finance, HR and Legal VLANs, using the VLAN IDs in Table 5.
- **DHCP:** Implemented per Section 5 — a local server on standalone LANs, and one shared server (on the Finance VLAN) with DHCP relay (ip helper-address) for the HR and Legal VLANs.
- **DNS:** Hierarchical/recursive-style resolution across DNS-1, DNS-RD and ISP-DNS as described in Section 6.1, rather than a single flat internal DNS server.

- **Redundancy:** CR1 and CR2 provide dual, equal/near-equal-cost OSPF paths into the Data Center (R11) and R&D (R5), and dual paths from the core into HQ (R4), satisfying the multiple-path requirement in at least two networks.
- **Security:** Console password (cisco), privileged EXEC password (class), and VTY/Telnet password (network) configured on every internal router; each router is named with the designer's first name and an index (Shrine1–Shrine10).
- **Default Route:** A static default route is configured on BR toward the ISP and redistributed into OSPF via `default-information originate`, so every internal router learns 0.0.0.0/0 without additional route entries. On the ISP side, exactly one static route (10.0.0.0/16 via BR) provides the return path, satisfying the "no dynamic routing from the ISP, minimum route entries" requirement.

8 Conclusion

This report documents the completed implementation of the KtmTechnologies enterprise mini-network in Cisco Packet Tracer. The three-area OSPF plan, VLSM addressing and point-to-point link plan were carried through into the final configuration and VLAN IDs, DHCP server placement/relay, and a hierarchical recursive-style DNS and web-hosting configuration were finalized to complete the design. The result is a three-area OSPF network covering all specified departments and the branch office, with default gateways, DHCP scopes and multi-level name resolution configured consistently across every LAN and VLAN, and redundant core paths demonstrating OSPF failover into at least two areas.

9 Project Repository

The Cisco Packet Tracer project file and related network-design information are available in the following GitHub repository:

`github.com/ShrineSigdel/Network_Design_in_CISCO_pkt_tracer`